Hellenic Complex Systems Laboratory

Network of Musical Instruments for Rhythm Accompaniment

Technical Report XV

Chrysavgi Chatzimichail and Aristides T. Hatjimihail

2018

Network of Musical Instruments for Rhythm Accompaniment

Chrysavgi Chatzimichail [a] and Aristides T. Hatjimihail [b]

[a] avgi@hcsl.com, [b] ath@hcsl.com,

[a,b] *Hellenic Complex Systems Laboratory*

Short Description of the Demonstration

This Demonstration plots a network encoding musical instruments used for rhythm accompaniment. The data consist of 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms. The user can select various measures. The results are also presented as tables and scatter plots.

Search Terms

network, graph, music, popular songs of Smyrna, musical instruments, rhythm accompaniment instruments, recordings

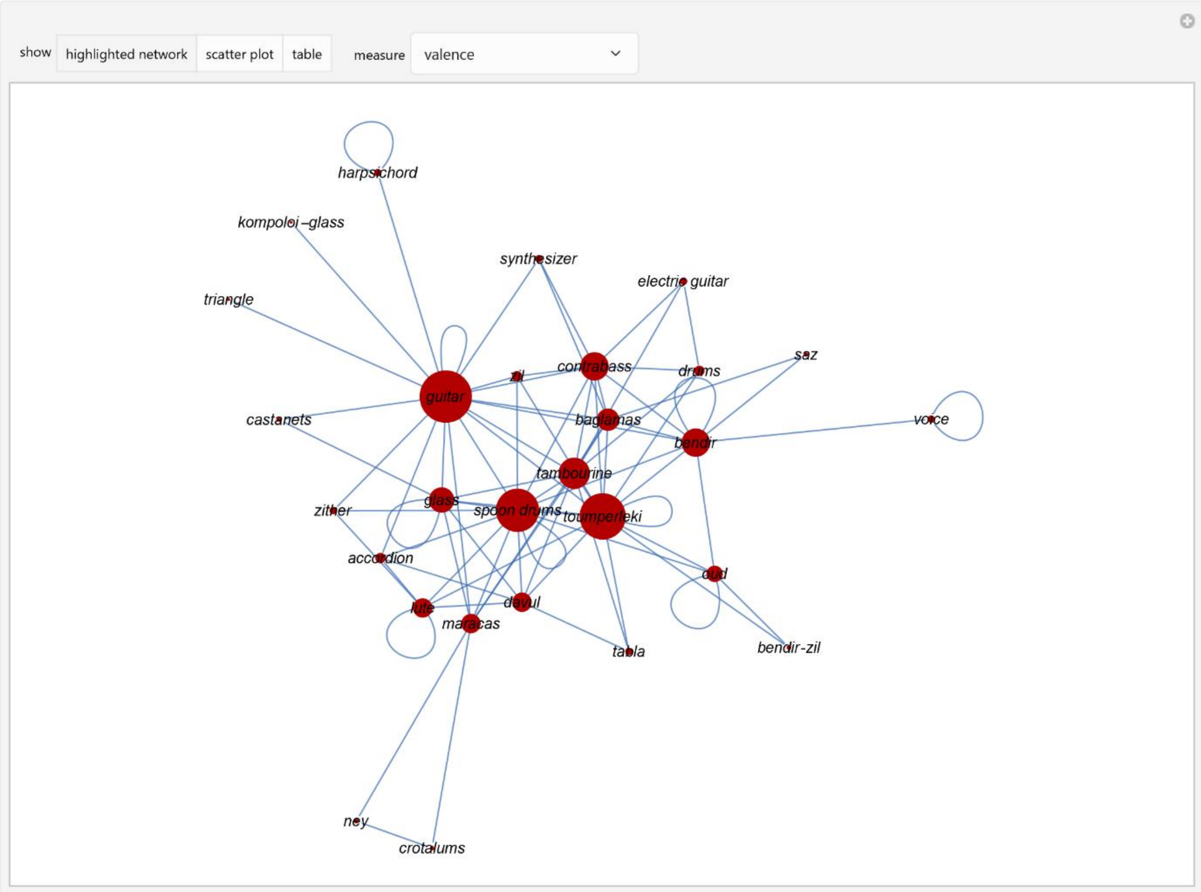


Figure 1: *A network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms. The area of each highlighted vertex is proportional to its valence.*

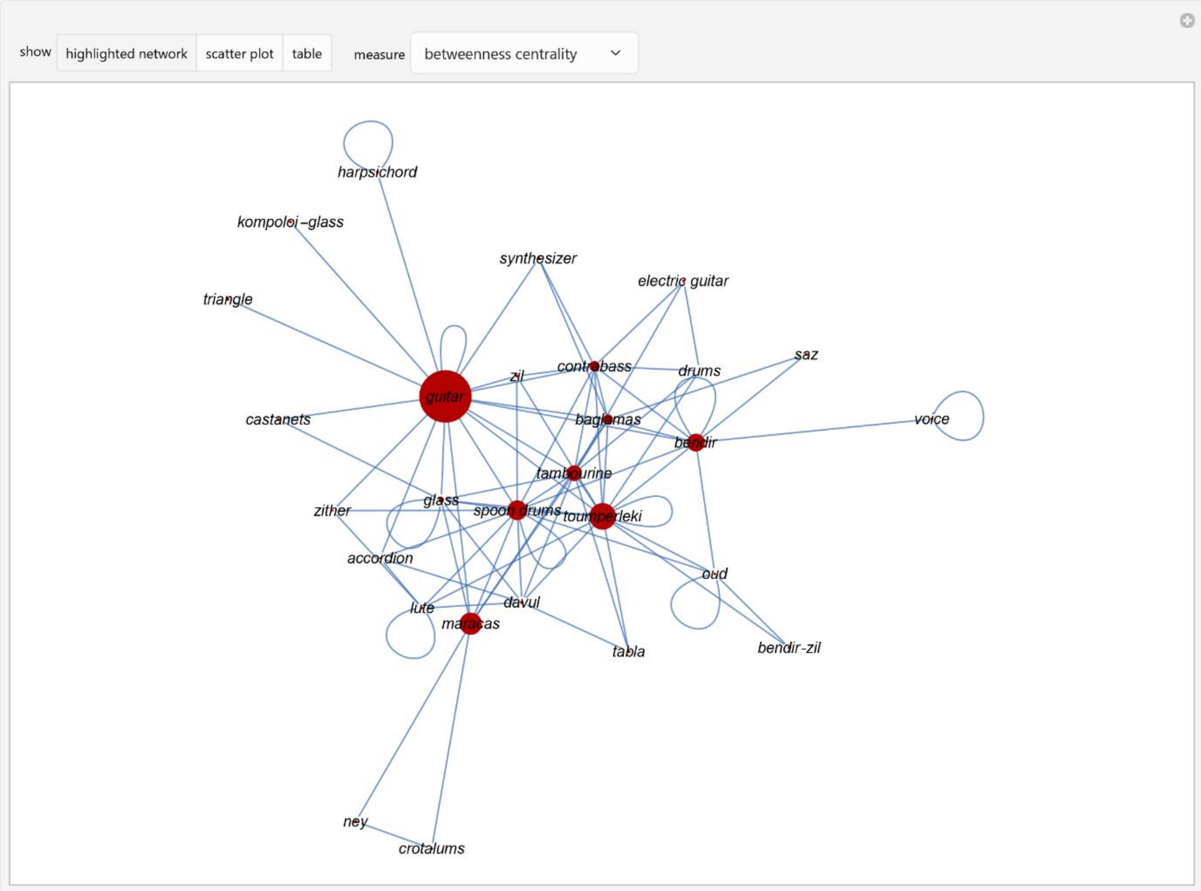


Figure 2: *A network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms. The area of each highlighted vertex is proportional to its betweenwss centrality.*

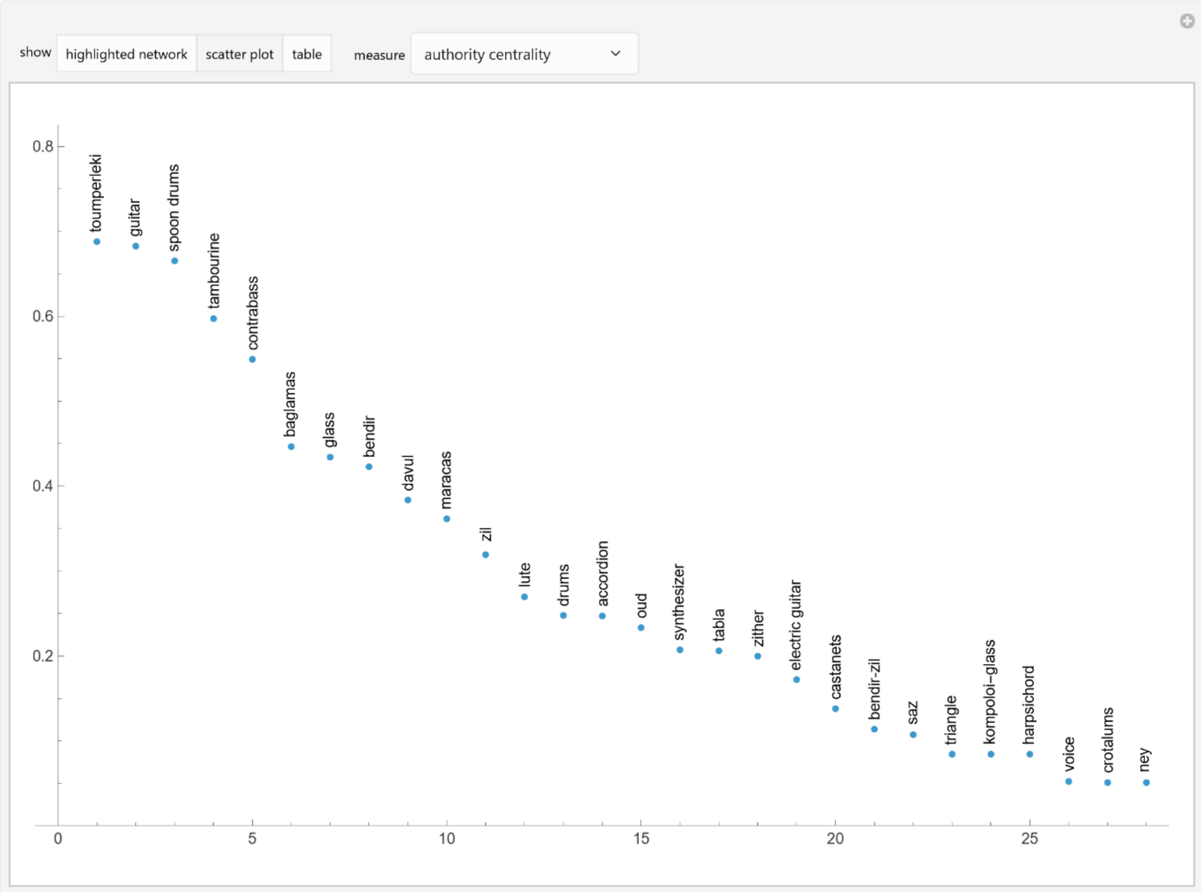
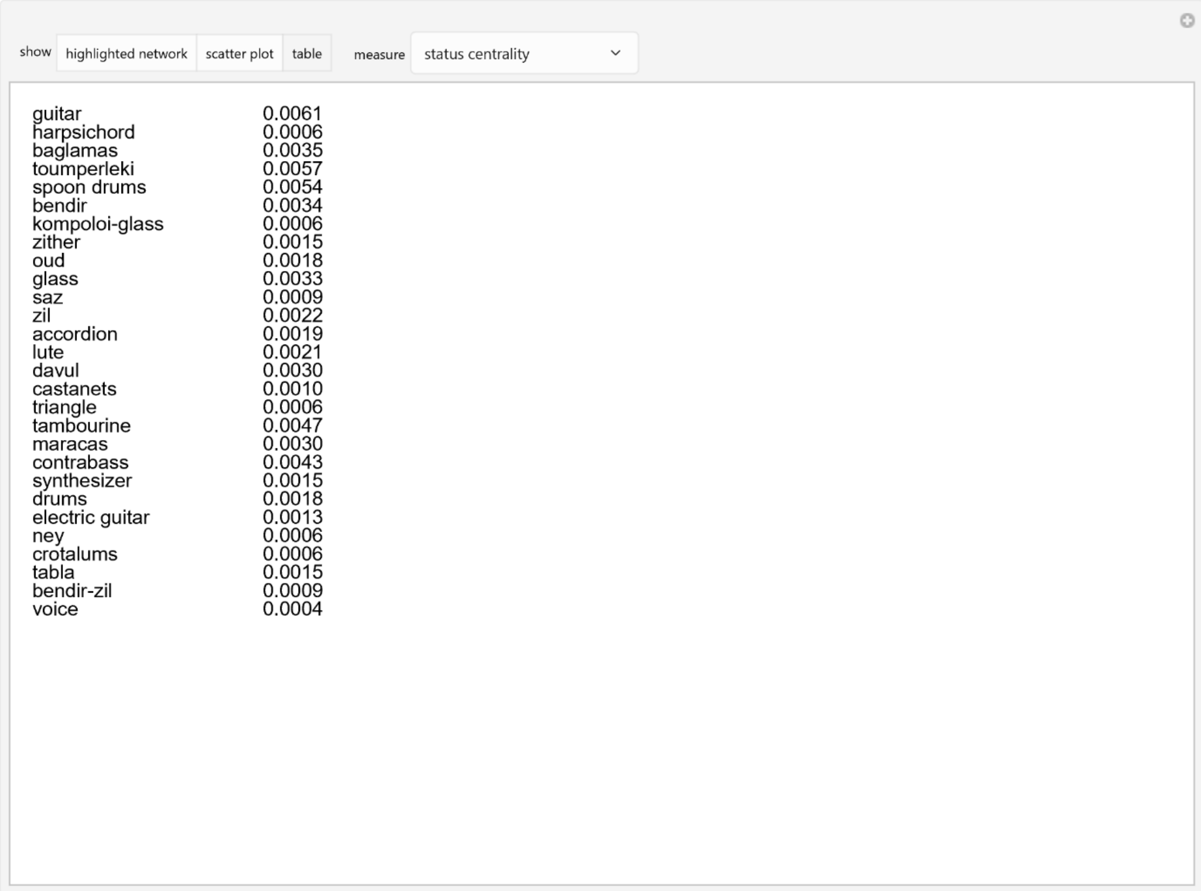


Figure 3: *A scatterplot of the authority centralities of a network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms.*



guitar	0.0061
harpsichord	0.0006
baglamas	0.0035
toumperleki	0.0057
spoon drums	0.0054
bendir	0.0034
kompoloi-glass	0.0006
zither	0.0015
oud	0.0018
glass	0.0033
saz	0.0009
zil	0.0022
accordion	0.0019
lute	0.0021
davul	0.0030
castanets	0.0010
triangle	0.0006
tambourine	0.0047
maracas	0.0030
contrabass	0.0043
synthesizer	0.0015
drums	0.0018
electric guitar	0.0013
ney	0.0006
crotalums	0.0006
tabla	0.0015
bendir-zil	0.0009
voice	0.0004

Figure 4: *A table of the status centralities of a network encoding musical instruments used for rhythm accompaniment in 100 recordings of 21 popular songs of Smyrna in nine-beat rhythms.*

Details

The following rhythm accompaniment musical instruments were considered: guitar, toumperleki, spoon drums, bendir, baglamas, contrabass, oud, harpsichord, glass, zil, tambourine, lute, maracas, davul, saz, zither, bendir with zil (bendir-zil), drums, castanets, accordion, kompoloi with a glass (kompoloi-glass), synthesizer, tabla, crotalums, ney, electric guitar, and triangle, and voice.

The network encodes the use of these musical instruments in the recordings, either alone or in combination. Each vertex of the network represents an instrument. If an instrument was used alone, it is connected to itself with a loop. If it was used in combination with any other instruments, it is connected to each of them with an edge. The network is weighted. The weight of each loop or edge is the frequency of use for each instrument or combination of instruments in the recordings. The area of each highlighted vertex is proportional to its respective measure. The calculated measures are the valences, the closeness, betweenness centrality, degree, radiality, eccentricity, hub, authority centrality, eigenvector centrality, status centrality, page rank centrality, local clustering coefficients and mean neighbor degrees.

To the best of our knowledge, this Demonstration presents a novel method for studying the characteristics of musical instruments.

Reference

[1] C. Chatzimichail, "The Popular Songs of Smyrna in Nine Beat Rhythms Before and After the Destruction of Smyrna," thesis, Department of Traditional Music, Technological Educational Institute of Epirus, Greece, 2017. https://doi.org/10.17605/OSF.IO/WEK3Q.

Available at: https://thesiscommons.org/wek3q/

Demonstration

DOI

10.5281/zenodo.20415355

Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/NetworkOfMusicalInstrumentsForRhythmAccompaniment.nb

First published in 2018 as part of the Wolfram Demonstrations Project:
https://demonstrations.wolfram.com/NetworkOfMusicalInstrumentsForRhythmAccompaniment/

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS
Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.
System memory (RAM): 4 GB+ recommended.

Citation

Chatzimichail C, Hatjimihail AT. *Network of Musical Instruments for Rhythm Accompaniment.* Ver. 1.2.1. Hellenic Complex Systems Laboratory; 2026. https://doi.org/10.5281/zenodo.20415355

License

Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Technical Report

DOI

10.5281/zenodo.20415370

Citation

Chatzimichail C, Hatjimihail AT. *Network of Musical Instruments for Rhythm Accompaniment.* Technical Report XV. Hellenic Complex Systems Laboratory; 2018. https://doi.org/10.5281/zenodo.20415370

License

Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

First Published: May 14, 2018

Revised: May 27, 2026